Amazon Sales Performance Analysis

Excel Case Study Documentation

Order data: 2 February 2025 - 2 April 2025

Orders analyzed	Total Sales	Categories	Products	Locations
250	243,845	5	10	10

Tool: Microsoft Excel (Tables, formulas, PivotTables, PivotCharts, Slicers)

1. Introduction

E-commerce sellers rely on data-driven decisions to manage sales, inventory and customer engagement. This case study analyzes order-level Amazon sales data in Microsoft Excel, covering data checks, calculated fields, PivotTable analysis and an interactive dashboard.

2. Business Problem

An Amazon seller wants to understand how its orders are performing and where attention is needed. The analysis answers:

- Which products generate the highest total sales?
- Which categories contribute the most to total sales?
- Which customer locations show strong or weak performance?
- Which payment methods are used most often?
- What share of orders are completed/delivered, pending, or cancelled?
- Which customers contribute the highest total sales?
- How does sales performance change by month?

3. Project Objectives

- Load and understand the dataset structure.
- Check data quality and standardize the data.
- Create calculated fields for analysis.
- Analyze sales using PivotTables.
- Build an interactive dashboard with KPIs, charts and slicers.
- Present insights and recommendations.

4. Dataset Description

The workbook contains **250 order records** from **10 customers** in **10 locations**.

Column	Description
Order ID	Unique identifier for each order
Date	Date the order was placed
Product	Product purchased
Category	Product category
Price	Price per unit
Quantity	Units ordered
Total Sales	Price x Quantity
Customer Name	Name of the customer
Customer Location	Customer city

Column	Description
Payment Method	PayPal, Credit Card, Debit Card, Gift Card, Amazon Pay
Status	Original order status (Completed, Pending, Cancelled)

Categories: Electronics, Home Appliances, Footwear, Clothing, Books.

Products: Refrigerator, Laptop, Smartphone, Washing Machine, Smartwatch, Headphones, Running Shoes, Jeans, T-Shirt, Book.

5. Methodology

5.1 Data Loading

- Imported the order data into Excel and reviewed the structure.
- Checked data types and formatting.

5.2 Data Checks

- **Duplicates:** none found (no duplicate rows or Order IDs).
- **Missing values:** none found.
- **Formats:** Date converted to short date format; Price formatted to two decimal places; Price, Quantity and Total Sales are numeric.
- **Consistency:** category, product and location names are consistent.
- **Validation:** Total Sales = Price x Quantity for all 250 orders.

5.3 Calculated Fields

- **Month:** derived from the order date (Feb, Mar, Apr).
- **Order Status Flag:** groups the source status into Delivered (from Completed), Pending and Cancelled.

The dataset has no cost price or profit information, so the analysis focuses on Total Sales and order-level performance, not profit.

Important: Total Sales is the value of all orders placed, including pending and cancelled orders. It is not realised revenue.

5.4 PivotTable Analysis

Seven PivotTables summarize category sales, product sales, location sales, payment method usage, order status, top customers and monthly sales.

5.5 Dashboard

The Dashboard sheet contains Total Sales and Total Orders KPI cards, seven charts (Sales by Category, Sales by Product, Sales by Customer Location, Order Status Breakdown, Orders by Payment Method, Top Customers by Total Sales, Monthly Sales Trend) and four slicers (Category, Customer Location, Order Status, Month).

5.6 Workbook Structure

Sheet	Contents
Raw Data	Original order-level dataset
Cleaned Data	Dataset plus Month and Order Status Flag
Pivot Tables	7 PivotTables used for the analysis
Dashboard	KPIs, 7 charts and 4 slicers

6. Key Metrics

KPI	Result
Total Orders	250
Total Sales (all orders)	243,845
Completed / Delivered Orders	88 (35.2%)
Pending Orders	85 (34.0%)
Cancelled Orders	77 (30.8%)

The source Status field uses 'Completed'; the cleaned data maps it to the label 'Delivered'.

7. Insights and Findings

7.1 Electronics is the highest-selling category

Category	Total Sales	Share
Electronics	129,950	53.29%
Home Appliances	105,000	43.06%
Footwear	4,320	1.77%
Clothing	3,540	1.45%
Books	1,035	0.42%
Total	**243,845**	**100%**

Electronics and Home Appliances together contribute about 96.35% of total sales, so the business depends heavily on high-value products.

7.2 Refrigerator is the top-selling product

Rank	Product	Total Sales
1	Refrigerator	78,000
2	Laptop	58,400
3	Smartphone	48,500
4	Washing Machine	27,000
5	Smartwatch	15,750
6	Headphones	7,300
7	Running Shoes	4,320
8	Jeans	2,480
9	T-Shirt	1,060
10	Book	1,035

Refrigerator alone contributes about 31.99% of total sales. Refrigerator, Laptop and Smartphone together generate 184,900 (75.83%).

7.3 Miami leads locations, but sales are fairly evenly spread

Rank	Location	Total Sales	Share
1	Miami	31,700	13.00%
2	Denver	29,785	12.21%
3	Houston	28,390	11.64%
4	Dallas	27,145	11.13%
5	Seattle	26,890	11.03%
6	Boston	26,170	10.73%
7	Chicago	20,810	8.53%
8	New York	18,940	7.77%
9	Los Angeles	17,820	7.31%
10	San Francisco	16,195	6.64%

Miami is the strongest location, but its lead over Denver is only about 1,915. The weakest locations are San Francisco, Los Angeles and New York, each below 8% of total sales.

7.4 Only about a third of orders are completed/delivered

Status	Orders	% of Orders	Sales Value	% of Sales
Delivered / Completed	88	35.2%	88,530	36.3%
Pending	85	34.0%	90,285	37.0%
Cancelled	77	30.8%	65,030	26.7%
Total	**250**	**100%**	**243,845**	**100%**

- Delivered is the largest status group, but only narrowly (88 vs 85 pending).
- 64.8% of orders are pending or cancelled.
- Cancelled orders account for 65,030 of the total; excluding them, sales are 178,815.
- Delivered orders account for only 88,530 in sales.

7.5 PayPal is the most-used payment method

Payment Method	Orders	Percentage
PayPal	60	24.0%
Credit Card	54	21.6%
Debit Card	53	21.2%
Gift Card	42	16.8%
Amazon Pay	41	16.4%
Total	**250**	**100%**

Usage is spread across all five methods; no single method dominates.

7.6 February recorded the highest monthly sales

Month	Orders	Total Sales
February	113	122,695
March	131	117,730
April	6	3,420
Total	**250**	**243,845**

February's sales are slightly higher than March's even though March had more orders. The data starts on 2 February and ends on 2 April, so April contains only 6 orders and should not be compared with the full months.

7.7 Top customers by total sales

Rank	Customer	Total Sales
1	Olivia Wilson	36,170
2	Jane Smith	31,185
3	Emma Clark	29,700
4	John Doe	26,870
5	Emily Johnson	23,475

Olivia Wilson is the highest-value customer, based on total order value (including cancelled and pending orders).

8. Recommendations

- **Focus on high-performing categories.** Electronics and Home Appliances generate the majority of sales; prioritize them in inventory planning and promotions.
- **Keep top products in stock.** Refrigerators, Laptops and Smartphones are the strongest contributors; stock-outs here would cost the most.
- **Investigate pending orders.** 85 orders (34.0%, worth 90,285) are pending. Check payment issues, fulfilment delays, inventory availability and processing delays.
- **Reduce cancellations.** 77 orders (30.8%, worth 65,030) were cancelled. The dataset has no cancellation reason, so this needs additional data.
- **Strengthen strong locations and review weak ones.** Miami, Denver and Houston suit retention campaigns; San Francisco, Los Angeles and New York may need local promotions or investigation.
- **Keep multiple payment options.** PayPal is most used, but Credit Card and Debit Card are close behind.
- **Monitor monthly trends as more data arrives.** The data covers about two months with April nearly empty, so it is too short for reliable trends.

9. Limitations

- No cost or profit data, so profitability cannot be analyzed.
- Total Sales includes pending and cancelled orders, so it overstates realised revenue.
- The data covers about two months, and April has only 6 orders.
- Only 10 customers appear in the data, so customer rankings reflect a very small base.
- No cancellation or delay reasons are recorded.

10. Conclusion

The analysis covers **250 orders and 243,845 in total sales** from February to early April 2025. Electronics was the strongest category, Refrigerator the top product, Miami the top location and PayPal the most-used payment method. The most important finding is operational: only 35.2% of orders were completed/delivered, while 34.0% were pending and 30.8% cancelled. Because Total Sales includes all of these orders, addressing pending and cancelled orders is the clearest opportunity to improve results.